TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Dyes

A decision tree of solubility, affinity and redox tests that separates the application classes — with the reasoning chain an examiner expects for an unknown.

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 4 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to identify an unknown dye (or confirm a delivery) by its chemical behaviour: solubility, fibre affinity, and diagnostic reactions that separate the application classes. Identification runs on two specimens — the dye **powder** and a **dyed fabric** of unknown history. The logic is a decision tree: each test eliminates whole classes because classes differ in charge, solubility and redox behaviour (Theory Topics 2–3).

2. Test battery — dye powder

TEST	OBSERVATION	INTERPRETATION
Water solubility	Insoluble coloured powder, disperses only	Disperse, vat, sulphur or pigment
Soluble; dye a bleached cotton strip in neutral bath + salt	Cotton dyes directly	Direct or reactive (or solubilised vat)
Add alkali to the cotton bath, then soap hard	Colour survives boiling soap	Reactive (covalently fixed); direct bleeds heavily
Wool strip in acid bath	Wool dyes from acid	Acid or mordant dye
Acrylic strip	Brilliant dyeing from weakly acid bath	Basic dye
Hydrosulphite + NaOH on powder solution	Colour discharged, returns in air	Vat (reversible leuco); azo dyes discharge permanently if destroyed
Concentrated H_2SO_4 colour	Characteristic colour per class (e.g. many vat dyes give distinct	Confirmatory, class-level evidence

acid colours)

3. Test battery — dyed fabric (burning and stripping logic)

1. **Fibre first:** a dye is identified within its fibre context — burning/microscopy from earlier courses establishes cotton, wool, polyester or acrylic.
2. **Stripping tests on small cuttings:** (a) boiling water/soap — direct dyes bleed at once; (b) hydrosulphite/NaOH — vat and sulphur strip to leuco (vat re-oxidises in air); reactive resists stripping (covalent) though shade may dull; (c) hypochlorite destroys most classes but attacks fibre — last resort; (d) disperse dye on polyester strips into hot DMF solvent, colouring it — a near-specific test.
3. **Chromatography:** a dye extract spotted on filter paper/TLC and developed with water or butanol mixtures separates mixtures into bands — two dyes in a "single" shade reveal themselves.

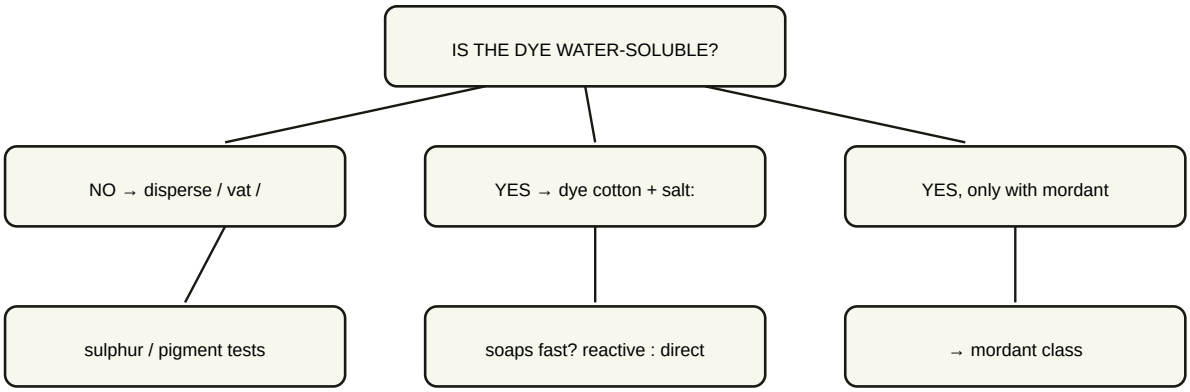


Fig. 1 — Decision tree for dye-class identification (powder route).

4. Worked identification (as recorded)

UNKNOWN U-2 — REASONING CHAIN

Powder soluble in water → not disperse/vat/pigment. Cotton + salt dyes strongly; after alkali treatment and hard soaping the shade persists with little loss → covalently fixed → **reactive dye**. Wool from acid bath dyes weakly (possible for many reactives), acrylic not dyed → basic excluded. Hydrosulphite discharges the solution irreversibly (azo chromophore destroyed) → azo-based reactive. Conclusion: CI class Reactive, azo chromophore. Each claim rests on one named observation — the format an examiner expects.

5. Precautions

- Concentrated H_2SO_4 tests: drop-scale only, acid into a dry tube, full eye protection; never on fabric worn or held.
- Run a known reference dye beside the unknown through the same tests — comparisons identify, absolutes mislead.
- Record negatives: "acrylic not dyed" excludes basic as firmly as a positive identifies.

6. Practice (viva preparation)

1. An unknown dyes polyester from a dispersion at 130 °C and strips into hot DMF. Class?
2. Two cotton dyes both dye with salt; one survives boiling soap after alkali, one does not. Classes and the bond difference?
3. Why is the hydrosulphite test done both on the solution and interpreted with re-exposure to air?
4. A dyed fabric bleeds in the first soap test yet is a reactive dyeing in production. Give the process explanation.

Answer key

1. Disperse dye — non-ionic, dispersion-applied, solvent-soluble; the DMF strip is characteristic.
2. Reactive (covalent bond to cellulose survives soap) versus direct (H-bonds/van der Waals desorb in hot soap).
3. Reduction alone is ambiguous — vat dyes and azo dyes both lose colour. Re-oxidation in air restoring colour proves a reversible leuco system (vat); permanent loss indicates chromophore cleavage (azo).
4. Unfixed hydrolysed reactive dye was not fully soaped off in production; the fibre-fixed dye is fast, the residue bleeds. It is a wash-off failure, not a class contradiction — re-soaping the fabric proves it.